

POINTE NOIRE, Congo

WMO: **644000**Lat: **4.816S**Lon: **11.887E**Elev: **17**StdP: **101.12**Time Zone: **1.00 (E01)**Period: **97-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	20.0	17.7	12.7	22.4	18.0	13.0	22.5	6.5	24.8	6.2	24.9	0.7	100	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.3	31.9	26.5	31.2	26.4	31.0	26.4	27.4	29.8	27.1	29.7	26.8	29.5	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	22.6	29.0	26.2	21.7	28.6	26.1	21.6	28.5	87.1	30.2	85.8	30.3	84.6	29.7	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.7	5.3	DB	17.8	32.8	0.8	0.6	17.2	33.3	16.7	33.6	16.2	34.0	15.6	34.4	(4)
(5)			WB	17.2	28.2	0.9	0.5	16.5	28.5	16.0	28.9	15.6	29.2	14.9	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	27.3	27.6	28.0	27.7	26.8	24.4	23.0	23.3	24.7	26.2	26.8	26.9	
	DBStd	1.91	0.93	0.96	0.90	0.89	1.10	1.08	0.95	0.89	0.86	0.76	0.77	0.84	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	5859	537	492	557	531	521	433	404	412	441	503	505	523	
	CDD18.3	2818	279	259	298	281	263	183	146	154	191	245	255	264	
	CDH23.3	23983	2900	2832	3394	3024	2403	904	380	393	899	1976	2392	2485	
	CDH26.7	5992	854	971	1261	991	545	56	6	5	41	230	477	554	
Wind	WSAvg	2.6	2.5	2.6	2.6	2.5	2.4	2.3	2.3	2.5	2.9	3.3	3.0	2.5	
Precipitation	PrecAvg	1238	175	211	204	144	51	1	1	2	14	73	205	152	
	PrecMax	1963	484	453	397	316	330	20	10	20	52	404	671	385	
	PrecMin	458	11	9	24	0	0	0	0	0	0	4	8	4	
	PrecStd	341	103	110	94	81	58	3	2	4	12	75	149	99	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	32.2	32.2	32.2	31.2	28.6	27.2	27.2	28.2	29.9	30.4	31.1	
		MCWB	26.5	26.5	26.5	26.6	26.4	24.1	22.4	22.2	23.6	25.2	25.7	26.0	
	2%	DB	31.1	31.6	31.9	31.5	30.2	28.0	26.3	26.3	27.8	29.1	30.0	30.2	
		MCWB	26.4	26.5	26.5	26.5	26.1	23.8	22.1	21.9	23.4	24.9	25.6	25.8	
	5%	DB	30.6	31.1	31.2	31.1	29.9	27.2	26.0	26.0	27.1	28.5	29.2	29.9	
		MCWB	26.2	26.5	26.4	26.5	26.0	23.3	21.9	21.8	23.0	24.6	25.2	25.7	
	10%	DB	30.0	30.8	31.0	30.3	29.1	26.9	25.2	25.2	26.2	28.0	29.0	29.1	
		MCWB	25.9	26.4	26.4	26.3	25.6	23.1	21.5	21.6	22.9	24.3	25.2	25.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	27.7	28.0	27.7	27.2	25.2	23.5	23.1	24.6	26.1	26.6	26.7
			MCDB	29.8	29.9	30.1	29.9	29.4	26.9	25.4	25.2	26.5	28.1	28.6	28.9
2%		WB	26.8	27.2	27.4	27.2	26.7	24.6	22.7	22.6	24.2	25.6	26.2	26.3	
		MCDB	29.5	29.8	29.9	29.7	28.8	26.4	25.0	24.8	26.1	27.6	28.3	28.6	
5%		WB	26.5	26.9	27.1	26.9	26.4	24.2	22.5	22.4	23.8	25.2	25.9	26.1	
		MCDB	29.2	29.8	29.8	29.5	28.5	26.1	24.7	24.5	25.7	27.2	28.1	28.4	
10%		WB	26.2	26.6	26.7	26.7	26.1	23.7	22.2	22.1	23.5	24.9	25.7	25.9	
		MCDB	28.8	29.4	29.5	29.2	28.2	25.7	24.3	24.2	25.5	26.9	27.8	28.2	
Mean Daily Temperature Range		5% DB	MDBR	5.5	6.0	6.3	5.9	5.0	4.9	5.1	4.7	4.1	4.0	4.6	5.0
			MCDBR	6.1	6.6	6.6	6.4	5.5	5.2	5.7	5.2	4.6	4.5	5.1	5.7
	MCWBR		2.5	2.9	2.8	2.7	2.2	2.0	2.2	1.9	1.7	1.8	2.1	2.3	
	5% WB	MCDBR	5.7	6.2	6.2	6.0	5.2	4.7	5.1	4.6	4.0	4.2	4.9	5.3	
		MCWBR	2.6	3.0	2.9	2.8	2.3	2.1	2.3	2.0	1.7	2.0	2.2	2.4	
Clear-Sky Solar Irradiance	taub	0.561	0.564	0.525	0.475	0.577	0.665	0.710	0.731	0.641	0.532	0.485	0.538		
	taud	2.031	2.030	2.155	2.319	1.963	1.737	1.640	1.617	1.807	2.120	2.273	2.098		
	Ebn at Noon	793	793	814	832	719	637	616	628	714	809	851	808		
	Edh at Noon	182	183	160	131	178	219	244	260	223	165	142	169		
All-Sky Solar Radiation	RadAvg	4.69	4.68	5.09	4.82	4.11	3.69	3.60	3.60	3.51	3.78	4.41	4.29		
	RadStd	0.24	0.31	0.27	0.25	0.33	0.27	0.23	0.29	0.17	0.17	0.28	0.32		